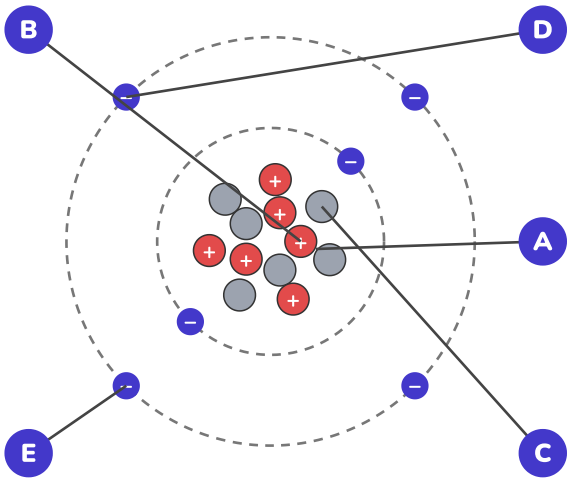
PARTS OF AN ATOM

Name: ..... **Date:** .....

Part 1: use the diagram to complete the table. Part 2: fill in the gaps in the particle table. Part 3: complete the sentences.

PART 1: NAME THE PARTS



This is an atom of carbon-12: atomic number 6, mass number 12.

Red particles carry a positive charge, grey particles no charge, and the small dark particles on the rings a negative charge.

Not to scale: a real nucleus is about 10 000 times smaller than the whole atom.

Letter	Name of the part	Charge (+, – or none)	Where is it?
A			
B			
C			
D			
E			

PART 2: THE THREE SUBATOMIC PARTICLES

Particle	Relative charge	Relative mass	Location
Proton		1	
Neutron	0		nucleus
Electron			

PART 3: COMPLETE THE SENTENCES

1. The atomic number of an element is the number of _____ in one atom.
2. In a neutral atom the number of electrons equals the number of _____.
3. The mass number is the number of protons plus the number of _____.
4. Almost all the mass of an atom is in its _____.

For non-commercial, education use only

PARTS OF AN ATOM ANSWER KEY

Name: ..... **Date:** .....

PART 1: NAME THE PARTS

A = Nucleus; B = Proton; C = Neutron; D = Electron; E = Electron shell (energy level)

A | Nucleus | positive overall | the centre of the atom

B | Proton | + | in the nucleus

C | Neutron | none | in the nucleus

D | Electron | – | moving around the nucleus in a shell

E | Electron shell (energy level) | not a particle | a ring around the nucleus where electrons are found

PART 2: THE THREE SUBATOMIC PARTICLES

Proton | +1 | 1 | nucleus

Neutron | 0 | 1 | nucleus

Electron | –1 | almost 0 (about 1/1840) | shells around the nucleus

PART 3: COMPLETE THE SENTENCES

1. protons 2. protons 3. neutrons 4. nucleus